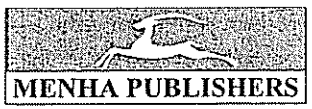
A Way with Words: Recent Advances in Lexical Theory and Analysis

A Festschrift for Patrick Hanks

edited by

Gilles-Maurice de Schryver



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Defining the Definiendum

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[*This paper is unfinished, with the last revisions dating from the end of February 2007; John Sinclair died on 13th March 2007.*]

The title of this paper seems on the surface to be the main task of lexicography; defining the definiendum – that's what we do. One of the early jobs in dictionary design is to establish a list of definienda, or headwords; this shapes the work, its size, scope and specialisation. But there is another sense of *define* – to specify the limits of something, the boundaries, rather than the meaning. The definiendum, the object of lexicographical activity, is no longer a simple entity, a headword; for the last twenty years evidence has been growing to suggest that the word is not the principal unit of meaning in a language.[1] The tendency of words to occur together more often than their frequency would predict was one of the first novel findings of research on large text corpora, and its implications are far-reaching in lexicography because the dominance of the word has been unchallenged for centuries.

Idiomatic phrases have been incorporated into dictionaries for a long time because of the obvious need for their own definitions. They tend to appear at the end of dictionary entries, and their citation forms (in bold face usually) are ad hoc and can vary from one publication to another. The evidence from corpora adds up to a strong case for extending the treatment of multi-word units of meaning – a much wider concept than idiom – and giving them the same status as the usual headwords.

This is a far-reaching change, because it affects the relationship between grammar and lexis. When the definiendum was a single word, it also had a clear place in the grammar and there was no conflict between the two aspects of language form. When the definiendum is elaborated to more than one word, com-

petition arises between grammar and lexis. To which party does the structure of this unit belong?

I will follow my usual practice here and use the term *lexical item* for any configuration of text which has a distinct sense, whether it consists of one or more words and whether or not those words are contiguous (Sinclair 1966, 1996, 1998). The lexical item in a dictionary is the definiendum.

For the purpose of presenting an argument here, it is necessary to represent grammar and lexis initially as competitive, but corpus study is gradually making available a more holistic approach, via phraseology, which does not demand initial separation of the two. It will depend on whether or not grammarians will give ground and recognise that there are linguistic structures which are not grammatical.

Example: *sever*

Let us take an example and consider the issues. The form *sever* occurs 599 times in The Bank of English after the proper name *Sever* has been removed. It is generally regarded as having a "core" sense, "divide by cutting or slicing" with a subsense "put an end to [a connection or relationship]; break off" (this is the wording of *The New Oxford Dictionary of English* (Pearsall, Hanks, *et al.* 1998)).

Digression: core sense

First, a short digression, because otherwise the reader may think that the core sense is somehow the important one, and the other is a minor usage. The situation is precisely the opposite. From a selection made arithmetically from the full concordance, with 81 instances, only 9 were of the core sense, and all the others were of the "put an end to" sense. There are a few paragraphs of justification in the Introduction to *The New Oxford Dictionary of English* (pp. ix-x), essentially defending long-established practice, but, strangely, also claiming support from modern corpora. The above analysis shows that there is very little support indeed from The Bank of English for the designated core sense to be a "typical, central use of the word in question". The sense of *sever* that is identified as the core is a rather specialised sense from the domains of cable-laying, accidents, gardening and surgery. Further bolstering for the traditional view is found in the claim that native speakers accept that the most literal sense is the core one. If they do (and it is not clear if they have been consulted) it is probably because they are familiar with dictionaries.[2]

Coselections with *sever*

Returning to the main theme, we see that *sever* is only one component of the sense of "put an end to" There are several other components, of which we describe the least abstract first.

The preposition *with*, very occasionally *to*, *between* or *from* occurs often to the right of *sever*, but is not the immediately following word. Around half of the instances show this, and we shall return to the others later, when the lexical item is more fully specified. In grammar these are prepositions, but in lexis they are just words. We must consider what it means to be a preposition.

Digression: preposition

A preposition is a grammatical item, and so is a member of a closed class. Members of a closed class make meaning by their contrasts, by the fact that in choosing one the others are rejected. This is one of the principal differences between grammar and lexis; in the latter there is no practical way of setting up closed classes. A closed class is usually small, and can be enumerated, and here the class of preposition is untypical, because large. Quirk *et al.* (1985: 665ff) list 72 single-word prepositions and a sample of two- and three-word ones. Whereas in choosing, say, the pronoun *we*, it is reasonable to explain it as simultaneously a non-choice of *he*, *she*, *you* etc., this is not satisfying when it comes to prepositions. On the other hand, prepositions do not constitute an open class like common nouns or even adverbs; these are much larger and are not enumerable.

So the prepositions overlap grammar and lexis; this is confirmed by the number of lexical items, like the one around *sever*, which select only a few from the list. Some affinity of sense can also be seen between the preposition chosen and what in grammar would be its object. In contrast, central grammatical classes and systems do not have selectional priorities, nor this kind of sense affinity. They are chosen on the basis of which one from the set is required, either by the needs of identification etc. (like *him* vs. *her*) or constraints like concord.

Colligation and colligation-like patterns

Firth (1968: 181ff) pointed out that as well as lexical collocation in text there were patterns of co-occurrence of grammatical classes. He called this *colligation*. When concordances became available, they were inevitably word-based,

because that was the best that the computers of the day could manage. Early study of the data showed that often a word and a whole grammatical class co-selected each other; so the word *lap* in its quasi-anatomical sense requires to be preceded by (a) a preposition, and (b) a possessive adjective such as *her* (Sinclair 2003: 73ff). The term colligation was extended to cover such cases.

In the case of *sever*, it is the choice of preposition that we must turn to. If many different prepositions are found here, and we see no reason why the others should not occur as well, then the occurrence of one of them is an instance of colligation. On the other hand, if only one appears, then it is a lexical selection. If, as often happens, one preposition dominates the frequencies but in certain circumstances one or two more are found, then we have a new situation. We have a variable in the specification of a lexical item.

One preposition dominates the concordance to *sever* – *with*. Infrequent alternatives are *to* and *between*. A few instances have *from*, e.g. *from each other*. We can set up a class for them, a small selection from the whole class of prepositions, those that can be selected to realise a component of the lexical item we are building up. This is not colligation; there is no grammatical imperative to make such a selection, and there are probably thousands of such selections in other lexical items, all at least slightly different. Nor is it lexicogrammar, which is grammar that is slightly more sensitive to lexis than pre-corpus grammars. It is lexical.

We shall make proposals later for how to build this into an appropriate descriptive model.

Semantic preference

Sever is a transitive verb in grammar, and the concordance shows a number of nouns that occur as the head noun of the object phrase. In order of t-score significance, these are: *ties*, *links*, *link*, *relations*, *connections*, *connection*, a very small group that all have a similar sense. By no means all the words that share this sense occur here, because this is not a conceptual category; a thesaurus entry would be much more comprehensive.

Here is a second lexical category, quite similar to the prepositions above.

Colligation continued

In between the preposition and the noun there are many instances of grammatical words, as follows (in t-score sequence). The main ones are: *all*, *his*, *its*, *their*. Apart from *all*, this looks like colligation with the grammatical class of possessive adjective, and indeed all the other possible choices occur in small numbers. So in this area we have a collocation with *all*, and a colligation with the grammatical class. They are in slightly different positions, because there is the occasional occurrence of both, e.g. *all his*.

A lexical item

We can now set out the components of a lexical item whose core is *sever*, apart from the final component, the *semantic prosody*, which we will return to.

Core: *sever*
Collocations: (a) *with*, *to*, *between*, *from*
(b) *all*
Colligation: possessive adjective
Semantic preference: *ties*, *links*, *link*, *relations*, *connections*, *connection*

We must consider what kind of organisation this is. While there are clear grammatical relationships, it is not a fragment of ordinary English grammar. But outside the grammar we have only very crude ways of specifying relationships, even positions. So we must find or propose a new kind of organisation for the lexical item.

Fortunately, there is an appropriate kind of organisation already in existence, though not recognised in conventional grammars. This is the *local grammar* (Barnbrook & Sinclair 2001), which arose from the perceptions of Gross (1993) that no possible general grammar could adequately describe a corpus. Since then, local grammars have found several uses (for example, Allen 1998, Hunston & Sinclair 2000, Barnbrook 2002), and here is another one.

The claim here is that each lexical item which contains variables has its own, unique local grammar. Simple lexical items – single words – and fixed idioms do not need such a grammar because there is no variation. As far as possible the local grammar shares categories with the general grammar, but it reserves the right to establish its own categories where necessary.

The local grammar for *sever* so far is:

lexis:	*sever*	[*all*]	[POSSADJ]	SEMPREF	|*with* |*to* |*between* |*from*
grammar:	predicator	pre-deictic	deictic	head	preposition

Items in italics are actual words; in small capitals are classes; square bracketed items are optional; vertical lists are alternatives. SEMPREF calls up just the class that has been identified here. Note that the vertical lists are paradigms of a sort, but they do not contrast like grammatical ones – they are words which are closely related, alternatives that share most of their meaning with the other members of the class. Such notation should be familiar, because in recent years, in phraseology and in some modern grammars, strings of this kind are presented informally to show pattern.

This is a single coselection of up to five words, constituting a single lexical item. It is a joint construction of grammatical and lexical categories.

Alternative organisations

In the section above, "Coselections with *sever*" it was pointed out that around half of the occurrences of *sever* in this sense lacked a preposition. These we must now consider, to see if they can fit into the specification above or are doing something else.

1. There are a number of instances in which the kind of information found in the prepositional phrase is present, but an alternative construction is chosen, as in:

> because the holder of the office is supposed to sever all party links

> DAVID BECKHAM is preparing to sever his England ties!

The modifier *party* is roughly equivalent to "with the party". This is a regular relationship, often called nominalisation. The short form is not as explicit, but the meaning is easy to infer. So with *England*. In other instances the occurrence of words like *friendship* and *marriage* after *sever* identifies the ties satisfactorily. We can label these *pre-specified*.

1.1. A variant of prespecification is where cohesive reference points back to where the ties etc. are named:

> their only hope is to sever such links

2. Where there is no overt reference to the material we may expect in the prepositional phrase, some instances make it obvious that this information can be inferred.

> Initially, the club offered him £100,000 to work as a United ambassador for just 30 days a year. But Fergie was insulted and announced he would sever all ties.

The occurrence of *club* might allow this instance to qualify as a pre-specification, but it is a little remote. In headlines, however, inference is often essential:

> Gary's in a hurry to sever Links

3. In a few minor varieties of English, the lack of specification seems to be deliberate tactics, and we must label these *unspecified*. Here are two instances, one with a religious flavour and the other from a horoscope.

> This is the good we dare not sever.

> Ultimately, of course, we have no choice other than to sever our attachment and to build a life of our own, since the Uranus player cannot be possessed...

We can now complete the local grammar:

(a) specified

lexis:	*sever*	[*all*]	[POSSADJ]	SEMPREF	|*with* |*to* |*between* |*from*
grammar:	predicator	pre-deictic	deictic	head	preposition

(b) prespecified

lexis:		*sever*	[*all*]	[POSSADJ]	SEMPREF
grammar:		predicator	pre-deictic	deictic	head
paraphrase:	|nomin. |cohesion |word				

"nomin." stands for nominalisation, "word" for a word like *marriage* that specifies by itself.

(c) inferrable

lexis;	*sever*	[*all*]	[POSSADJ]	SEMPREF
grammar:	predicator	pre-deictic	deictic	head

Comments

There are two points to note from this description. One is that the scope of a local grammar is much wider than that of a general one, and handles material which relies on semantic interpretation. This indicates a further need for flexibility in grammars; when applied to corpus data they have to move beyond the tightly-enumerated classes. The need for this has been demonstrated on many occasions (for example, in the case of *brook*: see Sinclair 2003: 43ff.).

The other point is that the category of *paraphrase* is introduced to label the various kinds of prespecification. Because of the rich resources of a natural language, it is beyond ambition to imagine that a circumscribed description will capture all the options; paraphrase is thus a key category because it is not precise. This is not the place to discuss the role it may have in new descriptions, but it is likely to be of central importance.

Meaning-shift

At the outset of this description, the sense "put an end to" was distinguished from the cutting sense, and we have concentrated on it. There are three claims to be made:

(1) Text strings which conform to the local grammar are very likely to have this sense.
(2) Text strings which vary substantially from the local grammar are unlikely to have this sense.
(3) Each of the lexical components of the lexical item should show a meaning-shift from its use outside the lexical item.

The last claim needs some elucidation. The sense of *sever* has clearly moved away from physical cutting to the area of human and institutional relationships. With *ties* and *links* the shift is to select just only one of their potential senses – this is a most important function of the lexical item. There are several kinds of relationship, but the only relevant ones here are the institutional ones; the same goes for *connections*.

These shifts are quite obvious, but there are many in other items which are quite difficult to detect. Also the shift in sense of the prepositions may be too slight to be captured in a description – this is an open question at present; colligations are certainly exempt.

The multi-word lexical item is thus identified both by its conformity to its local grammar and the meaning-shift that it induces in its components.

Semantic prosody

The two obligatory elements of a lexical item are the core and the semantic prosody. So far we have deferred consideration of the semantic prosody, and indeed the focus of this example is primarily structural, whereas the semantic prosody, as its name suggests, is concerned with broad aspects of meaning, including attitudes, tactics and pragmatics. It tries to answer the question "for what reason was this particular item chosen to express the meaning, rather than a close alternative?" In the present case, there is a formal, bureaucratic aspect of this item that goes with its institutional use. There is also a greater feeling of precision and finality about the item than "put an end to" or "break off". "*Sever ... ties*" has the aura of the committee room, where cool and tough decisions are taken. In relation to marriage and more personal relationships, it highlights the more transactional or legal side of things.

Multi-word lexical items thus make available many shades of meaning that make possible effective and precise communication.

[*John Sinclair's draft paper ends at this point. By way of conclusion, a small random sample of corpus lines for* sever *is provided, drawn from the Bank of English. Apart from the addition of this data, the completion of references, and formatting, no changes have been made to the paper, which is published by kind permission of Elena Tognini-Bonelli.*]

Corpus sample: *sever*

```
 that period, it had been decided to sever all relations with two major
 since 1946, when islanders voted to sever all connections. Only after urgent
resigned as ALP president, needing to sever all political ties before moving
 ago. However, he must also agree to sever all links with Gucci. Gucci and
    to dismiss calls for Scotland to sever financial ties with England while
 worldviews because they threaten to sever her connections to her former life
 the new plan she would be forced to sever her links with all commercial
  might well be it." Monty would not sever his links with the game, but devote
<p> But Parlour now appears ready to sever his links with Highbury -if both
```

```
    had the cruel misfortune to also sever his cruciate ligament. <subh> Bonus
  </date> TIM Sherwood is expected to sever his ties with Blackburn later today
  ll be coming back. He might want to sever his connection formally," Mr Knight
 Unionist Party will almost certainly sever its institutionalised links with
 the views of those who would wish to sever our European Union connections.
  itself from other countries. 'If we sever our links # he said # we will not
  that his country does not intend to sever relations with the Soviet Union.
    Philippoussis, but he is loath to sever relations with such a class player
  asparagus. Use a saw-edged knife to sever spears a few inches below the
   of its sides equal in length. Now, sever the left hand side two inches from
     <p> The latest judgment will not sever the German state's close ties with
 reflexes because she did not totally sever the spinal cord. She has a scar on
    on the list; allied bombers would sever the power used to run refrigeration
 her son, and with them, might he not sever the attachment that binds them?
  Man for 70,000, a move was afoot to sever the ties. David Cannan, a former
 declarer naturally ducked his ace (to sever the defensive communications). He
     He called on all Arab nations to sever their economic and political ties
      of dollars, though they did not sever themselves from the union. I'm
   part of her religion with which to sever ties. Looking back some years
       into their cocoon you begin to sever ties with the outside because it is
 yet that authorities believe he will sever ties with rugby league. <p> Morgan,
 have come and gone. Radisson wants to sever ties with Tatum and favours the
         comfort and security, and to sever unwanted or costly ties. In this
```

Notes

[1] There is a large range of variation among the world's languages as to what exactly a word is and what it does, so this claim should be restricted to the more analytic languages such as English for the time being.

[2] There may be a conflict here between the commercial side of lexicography and the linguistic. If users really prefer a rare but concrete sense of a word to be the core, and the normal usage to be a sub-sense, then publishers will continue to design dictionaries in this way. But there is no basis for such a prioritisation in language description.

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